Fluid Structure Interaction
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Developing Methodology for Model Fitting to be used in Fluid-Structure Coupled Problem

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Abstract

This study develops a methodology for system identification in fluid-structure interaction (FSI) problems by integrating digital twin concepts with numerical simulations. Accurate modeling of structural behavior in multi-physics environments requires coupling governing equations with iterative refinement based on computed and measured data discrepancies. A simplified internal fluid flow over a slender beam in a channel is modeled using the CoSimulationApplication in Kratos Multiphysics, where three different Young's moduli are identified through velocity measurements and deformation sensors. The workflow involves extracting model values, computing displacements and velocities at sensor points, and formulating a response function to quantify errors between measured and computed statistical quantities. An optimization process iteratively refines material parameters to enhance system identification accuracy while assessing the impact of different coupling strategies on computational accuracy and efficiency. By leveraging numerical optimization and strong coupling techniques, this project enhances predictive modeling for digital twin applications, improving structural monitoring and control in multi-physics environments. The findings contribute to advancing numerical techniques for coupled problems, emphasizing stabilization methods, iterative solvers, and data-driven optimization to improve the accuracy and efficiency of FSI simulations.

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1 Introduction

The study of Fluid-Structure Interaction (FSI) is pivotal in understanding the complex interplay between fluid flow and structural response, a phenomenon that spans a wide range of natural and engineered systems. From biological processes, such as blood flow in arteries, to large-scale engineering applications like aircraft wings and bridges, FSI plays a critical role in ensuring the safety, efficiency, and performance of systems. This chapter provides an overview of FSI, its importance in engineering, the challenges associated with coupled simulations, and the computational approaches used to solve FSI problems. The discussion is structured to first introduce the fundamental concepts of FSI, followed by an exploration of its necessity in engineering, the challenges it presents, and the numerical methods employed to address these challenges.

1.1 Understanding Fluid-Structure Interaction and Coupled Problems

Fluid-Structure Interaction (FSI) describes the bidirectional interplay between fluid flow and structural response, a phenomenon crucial in both natural and engineered systems. From arterial blood flow influencing vascular walls to aerodynamic forces shaping aircraft wings and wind-induced vibrations affecting bridges, FSI governs numerous real-world applications where fluid forces drive structural motion and deformation. Capturing these interactions requires solving the governing equations for both domains simultaneously while ensuring proper information exchange at the interface. Traditional uncoupled simulations fail to capture these interdependencies, often leading to inaccurate predictions or oversimplified models. Instead, coupled FSI simulations, which integrate fluid and structural solvers, are necessary to account for transient deformations, dynamic loads, and geometric changes in real-time. However, these simulations pose significant computational challenges, including stability concerns, high numerical costs, and complex interface handling, requiring advanced methodologies for accurate and efficient modeling.

Coupled problems extend beyond FSI and are fundamental in multi-physics simulations where multiple interacting domains must be solved simultaneously. In aerospace applications, for instance, the aerodynamic loading on an aircraft wing alters its shape, which in turn modifies the surrounding airflow, affecting lift and drag forces in a continuous feedback loop. Similarly, in biomedical engineering, the motion of artificial heart valves is dictated by both blood flow dynamics and the structural compliance of the valve leaflets, necessitating a strongly coupled approach to ensure physiologically accurate simulations. Computational frameworks such as Kratos Multiphysics facilitate these coupled simulations, offering robust numerical methods to balance accuracy, efficiency, and stability. This study leverages these tools to develop an FSI model that integrates Finite Element and Finite Volume Methods, optimizing solver performance and improving predictive capabilities for real-world engineering applications.

1.2 Challenges in Coupled Simulations

Coupled simulations, particularly in FSI, present several computational and numerical challenges that make them significantly more complex than single-domain problems. One of the primary difficulties arises from the inherent nonlinearity of FSI systems. The interaction between fluid and structure leads to highly dynamic and transient behavior, requiring robust numerical techniques to ensure stability and accuracy.

A key challenge is the handling of interface conditions. The continuity of velocity and force equilibrium must be preserved at the interface between the fluid and structural domains. This requires special numerical techniques to ensure that information is transferred accurately and consistently between solvers. Improper handling of these conditions can lead to spurious oscillations, non-physical results, or even divergence of the simulation.

Another major challenge is numerical stability and convergence. FSI problems involve different time scales and stiffness ratios between fluids and solids, making stable convergence difficult to achieve. Explicit coupling schemes can lead to numerical instabilities unless relaxation techniques or adaptive time-stepping methods are implemented. On the other hand, monolithic approaches, which solve the coupled system simultaneously, often result in large, computationally expensive systems that require efficient solvers and preconditioners.

The choice of computational methods and discretization schemes also plays a critical role in addressing these challenges. Fluid flow is typically governed by the Navier-Stokes equations and solved using Finite Volume or Finite Element Methods, while structural deformation is modeled using elasticity theory within a Finite Element framework. The coupling of these discretization techniques demands careful implementation to avoid numerical artifacts and maintain accuracy at the interface.

Furthermore, computational cost and scalability remain significant concerns in FSI simulations. Large-scale simulations require High-Performance Computing (HPC) resources to handle the vast number of degrees of freedom present in both the fluid and structural domains. Efficient parallelization strategies, domain decomposition techniques, and optimized solvers are necessary to make coupled simulations feasible for real-world applications.

1.3 Applications of FSI in Engineering and Science

FSI plays a fundamental role in engineering and scientific advancements, governing the behavior of structures subjected to fluid forces. Accurate modeling of these interactions is critical for optimizing performance, ensuring structural integrity, and improving energy efficiency across multiple domains. In aerospace engineering, the dynamic response of aircraft wings to aerodynamic loads exemplifies the importance of FSI, as deformations in wing geometry alter airflow patterns, impacting lift and fuel efficiency. Similarly, jet engine turbine blades endure high-speed airflow, inducing vibrations that can lead to fatigue failure if not properly accounted for in design.

In civil and structural engineering, FSI is pivotal in assessing wind and seismic impacts on high-rise buildings and bridges. The catastrophic collapse of the Tacoma Narrows Bridge in 1940 highlighted the devastating effects of fluid-induced structural resonance, demonstrating the necessity of incorporating aerodynamic forces into modern infrastructure designs. In biomedical engineering, the pulsatile nature of blood flow interacting with arterial walls influences the progression of cardiovascular diseases. The design of artificial heart valves and stents relies on FSI simulations to ensure proper function and long-term durability in a dynamic biological environment.

Energy systems also depend on FSI to optimize efficiency and longevity. Wind turbines, for instance, experience fluctuating aerodynamic forces that influence their structural response, requiring advanced simulations to prevent fatigue failure while maximizing energy output. Similarly, hydroelectric turbines operate under continuous water-induced loading, necessitating robust FSI models to predict structural deformations and extend operational lifespan. In the automotive and marine industries, vehicle aerodynamics and ship hydrodynamics are heavily influenced by FSI. The design of fuel-efficient automobiles must account for airflow-induced drag forces, while ship hulls interacting with ocean waves require hydrodynamic analysis to optimize performance and stability.

The ability to model and control fluid-structure interactions is essential for advancing engineering designs across these fields. With increasing computational power, FSI simulations continue to drive innovation by enabling predictive modeling of highly complex, nonlinear interactions between structures and their surrounding fluid environments.

2 Theory

2.1 Governing Equations

The governing equations for FSI are derived from fluid dynamics and structural mechanics principles. The Navier-Stokes equations model the fluid domain, while linear elasticity governs the structural domain. The coupling between these domains is achieved through a co-simulation approach within the Kratos Multiphysics framework.

2.1.1 Fluid Dynamics

The fluid domain is governed by the incompressible Navier-Stokes equations, describing the conservation of momentum and mass:

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u}\right) = -\nabla p + \mu\nabla^2\mathbf{u} + \mathbf{f}, \tag{1}$$

$$\nabla\cdot\mathbf{u} = 0. \tag{2}$$

These equations are discretized using the Finite Element Method (FEM), with stabilization techniques such as the Variational Multiscale (VMS) method and Streamline Upwind Petrov-Galerkin (SUPG) to mitigate convective instabilities. For transient simulations, an implicit time integration scheme based on the Backward Differentiation Formula (BDF) ensures stability and accuracy.

2.1.2 Structural Mechanics

The structural domain follows the equations of linear elasticity, defining the dynamic response of a deformable solid under external loads:

$$\rho_s\frac{\partial^2\mathbf{d}}{\partial t^2} = \nabla\cdot\sigma + \mathbf{b}, \tag{3}$$

$$\sigma = \mathbf{C} : \varepsilon, \tag{4}$$

$$\varepsilon = \frac{1}{2}\left(\nabla\mathbf{d} + \nabla\mathbf{d}^T\right). \tag{5}$$

FEM is used for discretization, with implicit time integration schemes such as Newmark-beta or Generalized-alpha methods to manage dynamic effects and maintain numerical stability.

2.2 Computational Framework

The FSI problem in this study is solved using a partitioned approach via Kratos Multiphysics' CoSimulationApplication [1]. The coupling enforces velocity and displacement continuity at the interface to ensure force equilibrium. A Dirichlet-Neumann scheme is employed, where the fluid solver computes pressure and velocity fields, transferring them as Neumann conditions to the structural solver, which in turn applies Dirichlet conditions back to the fluid. This iterative Gauss-Seidel coupling refines exchanged values until convergence.

The fluid domain is governed by the incompressible Navier-Stokes equations, discretized using the Finite Volume Method (FVM) or FEM, with stabilization techniques such as Streamline Upwind Petrov-Galerkin (SUPG) and Variational Multiscale (VMS). Structural deformations are modeled using Finite Element Analysis (FEA) with implicit time integration schemes like Newmark-beta and Generalized-α to ensure stability in transient simulations.

Accurate data transfer across non-matching grids is achieved through interpolation-based mapping, while Lagrange multipliers and Mortar methods enhance constraint enforcement. Advanced coupling strategies such as Robin-Robin refinement improve stability for highly flexible structures. The strong and weak formulations for Gauss-Seidel coupling are given by:

$$\mathbf{x}^{(k+1)} = \mathbf{D}^{-1}(\mathbf{b} - \mathbf{L}\mathbf{x}^{(k+1)} - \mathbf{U}\mathbf{x}^{(k)}), \tag{6}$$

where $\mathbf{D}$ is the diagonal matrix, $\mathbf{L}$ is the lower triangular part, and $\mathbf{U}$ is the upper triangular part of the system matrix.

To address the high computational demands of FSI, Krylov subspace solvers such as GMRES and BiCGSTAB are used for sparse linear systems, with preconditioners like Incomplete LU (ILU) factorization and algebraic multigrid (AMG) accelerating convergence. Adaptive Mesh Refinement (AMR) dynamically allocates computational resources to critical regions, enhancing resolution without excessive cost. OpenMP-based parallel computing further optimizes efficiency by distributing workloads across processors.

By integrating these numerical techniques and coupling strategies, this framework ensures accurate and computationally efficient FSI simulations. The following sections provide further details on solver settings, interface handling, and validation procedures.

2.3 Optimization Framework

This study involves model fitting for system identification within the FSI framework. The slender beam in a channel is simulated with two different Young's moduli, identified by measuring fluid velocity at specific points and deformation sensors at given locations. The goal is to optimize the structural model based on sensor data by minimizing the error between measured and computed responses. The optimization function used for model fitting is given by:

$$J = \frac{1}{2} \sum_{t=1}^{N} \sum_{i=1}^{M} \left[h_i^c(t) - h_i^m(t)\right]^2, \tag{7}$$

where $h_i^c(t)$ represents the calculated response from the simulation, $h_i^m(t)$ represents the measured response, and the summations account for all the sensor locations and time steps. The error function minimizes the difference between the computed and measured sensor values by iterating the model parameters. Differential Evolution (DE) optimization technique is used to improve system identification accuracy.

2.4 Methodology

This study employs a computational framework for solving FSI problems using Kratos Multiphysics, integrating numerical modeling with optimization techniques for system identification. The fluid and structural domains are governed by the incompressible Navier-Stokes equations and linear elasticity theory, respectively, and are discretized using the FEM to ensure numerical stability and accuracy. A partitioned coupling strategy is adopted, enabling independent solvers to exchange boundary conditions iteratively through Dirichlet-Neumann coupling, where the fluid solver imposes velocity constraints on the structure, and the structural solver computes reaction forces influencing the fluid. Relaxation techniques are employed to enhance convergence and ensure stability in strongly coupled interactions.

The solution process leverages efficient iterative solvers like GMRES and BiCGSTAB, alongside preconditioning techniques such as ILU factorization and algebraic multigrid (AMG), to solve large-scale sparse systems. Parallel computing with OpenMP is utilized to further accelerate computations. An integrated optimization framework iteratively adjusts Young's modulus to minimize the error between simulated and measured structural responses, with the objective function defined as the squared error of displacements and velocities. DE is employed to navigate nonlinear, multi-modal search spaces effectively, refining material parameters until convergence criteria are satisfied.

This methodology combines advanced coupling strategies, solver optimizations, and robust optimization techniques to ensure accurate and efficient FSI simulations, enabling high-fidelity predictive modeling and improved system identification.

3 Experimental Set-up

3.1 Model description

The model consists of a wind tunnel with a total length of 1.75 m, within which a vertically oriented beam is positioned 0.5 m from the inlet. The beam has a height of 0.25 m, while the total height of the tunnel is 0.5 m.

To capture the fluid-structure interaction, two velocity sensors are placed within the tunnel to monitor velocity variations, and two displacement sensors are mounted on the beam to track its deformation. The objective of this model is to simulate the coupled behavior of the fluid and the beam using Kratos Multiphysics. By analyzing the sensor data, we can evaluate the relationship between the beam's deformation and the changes in the surrounding flow field. This serves as the foundation for developing an optimization function that determines the optimal Young's modulus ratio when the beam is eventually split in the later stages of the process.

3.2 Meshing & Pre-Processing

The incident beam was bisected into two equal halves Figure 2, designated as the top and bottom portions, through manipulation of the computational mesh using the Mok_CSM.mdpa code Figure 3. This process involved redefining element boundaries by reassigning nodes to the respective top and bottom regions. The modified mesh defined two distinct, contiguous regions corresponding to the split beam.

The sensor placement process involved three distinct steps, each addressing a specific aspect of the measurement setup.

1. Complete Setup (Fig 1): illustrates the initial setup of the problem, including the location of the wall inside the wind tunnel, displacement sensors location and representation of location of velocity sensors. This figure serves as the basis for understanding the sensor placement strategy.

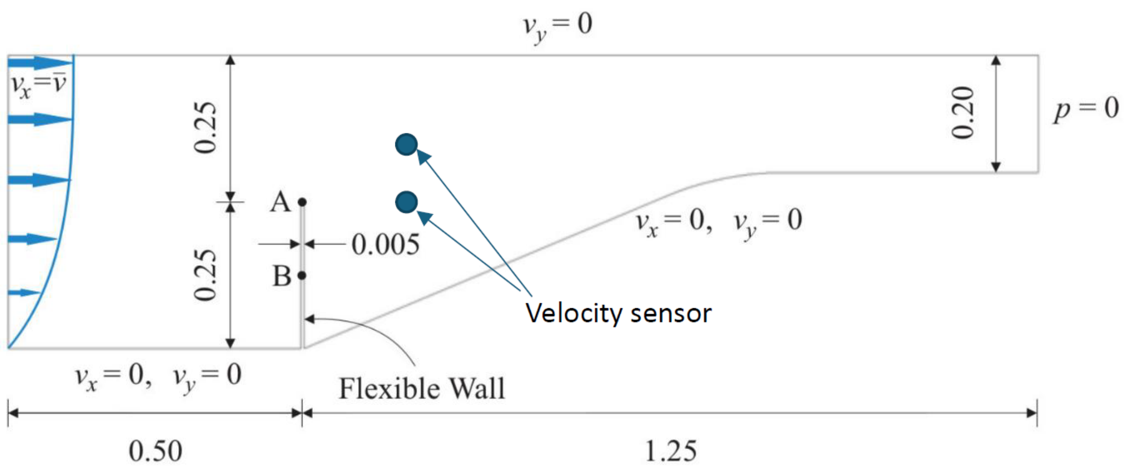

Figure 1: Complete Set-up

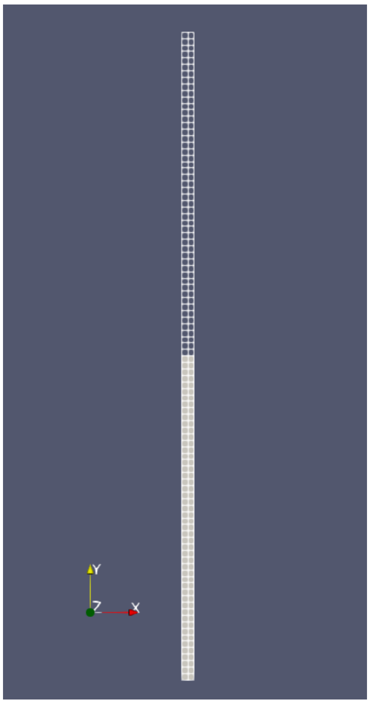

Figure 2: Mesh and split Details

```
//   SubModelPart: top_half
//     Number of Nodes: 154
//     Number of Elements: 100
//     Number of Conditions: 0
//     Number of Properties: 2
//     Number of SubModelParts: 0
//   SubModelPart: bottom_half
//     Number of Nodes: 150
//     Number of Elements: 100
//     Number of Conditions: 0
//     Number of Properties: 2
//     Number of SubModelParts: 0
```

Figure 3: Mesh Code Manipulation

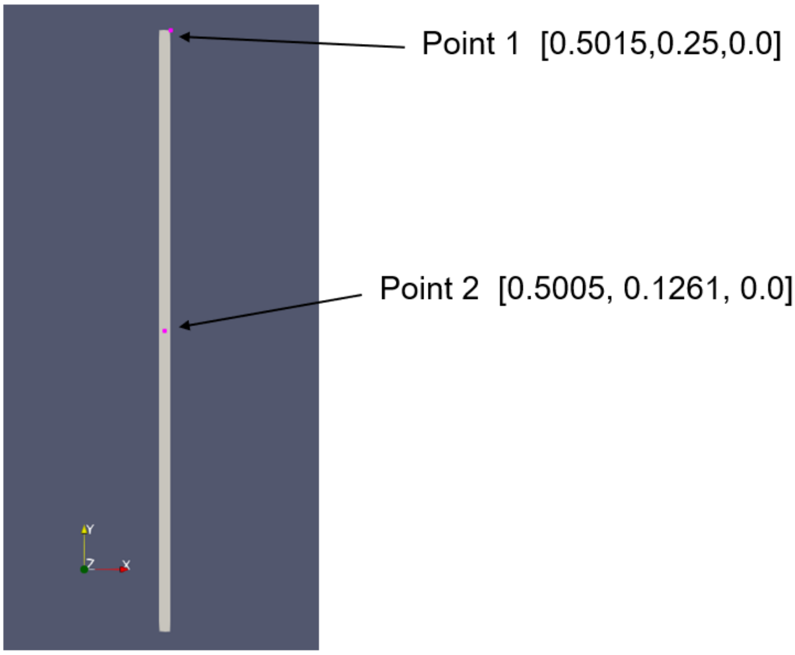

Figure 4: Location of Displacement Sensors

2. Displacement Sensors (Fig 4): The displacement sensors were pre-placed at locations A and B. The task was to identify the corresponding nodes/elements within the computational mesh that represented these locations. This was achieved by comparing the physical coordinates of A and B with the nodal coordinates in the mesh to locate the points and identify the corresponding.

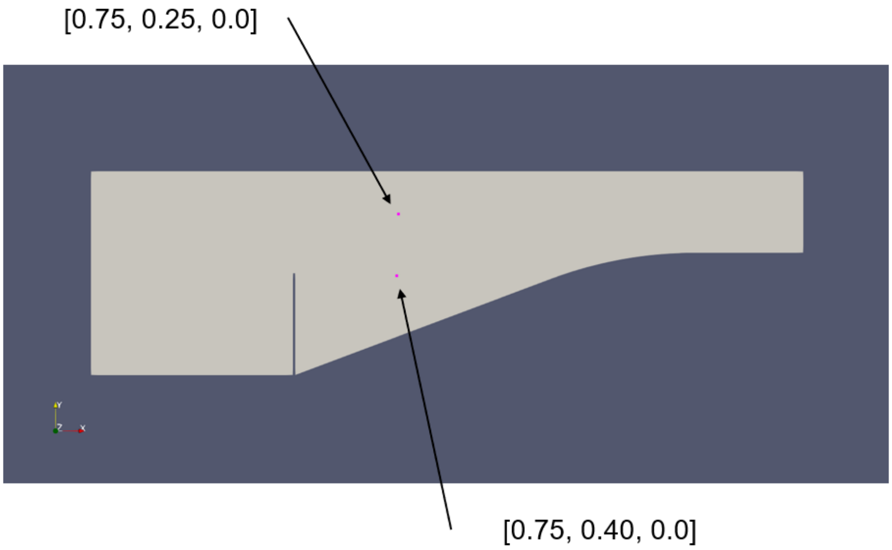

Figure 5: Location of Velocity Sensors

3. Velocity Sensors (Fig 5): Figure 1 provided a general indication of the desired velocity sensor locations. To determine their exact coordinates, we employed a combination of mathematical estimations and visual analysis. The sensors were then positioned accordingly at the calculated locations. Figure 5 shows the final sensor positions.

4 Simulation Parameters

This section describe the type of fluid and solid solver setting and the coupling strategy used.

4.1 Fluid Solver Settings

The fluid solver in this model employs an Arbitrary Lagrangian-Eulerian (ALE) formulation, a widely used method for handling mesh motion in fluid-structure interaction problems. The ALE approach allows for smooth deformation of the computational domain, which is critical for accurately resolving the interface dynamics between the fluid and the flexible structure. The solver adopts a monolithic approach for solving the Navier-Stokes equations, which inherently couples velocity and pressure in a single system, improving numerical stability and robustness compared to fractional step methods.
The mesh motion solver utilizes the structural similarity method, ensuring that mesh displacement mimics structural deformation in a stable and smooth manner. This technique prevents excessive mesh distortion and maintains the quality of elements in the computational domain. The boundary movement is constrained to preserve numerical stability, and mesh motion is explicitly handled for the interface, where strong coupling with the structural domain occurs.
The time stepping follows a fixed increment of 0.1 seconds, without automatic adaptation. While adaptive time-stepping can enhance efficiency in rapidly changing flow conditions, the chosen fixed time step ensures consistency and simplifies the implementation of coupling iterations with the structural solver.
For turbulence modeling and numerical stabilization, the solver adopts a Variational Multi-Scale (VMS) formulation, which is effective in capturing small-scale turbulent structures within the flow. The dynamic tau parameter is set to 1.0, balancing numerical dissipation and stability across the varying flow conditions. Convergence is strictly enforced with absolute and relative tolerances of 10^{-6} for both velocity and pressure to ensure high accuracy in the solution. A maximum of 10 iterations per time step is allowed for

the fluid solver to converge, ensuring efficient computation while maintaining stability.
The boundary conditions are carefully assigned to ensure realistic flow behavior. A time-dependent velocity profile is imposed at the inlet, with a no-slip boundary condition applied at the bottom wall and the interface. A slip condition is enforced at the top boundary to permit tangential movement while preventing normal penetration. At the outlet, a fixed pressure of zero is prescribed to allow smooth outflow and prevent numerical instabilities.
Mesh displacement is fully constrained along all domain boundaries, except at the interface, to maintain numerical stability. The deformation of the computational grid follows the structural motion to ensure accurate resolution of interface dynamics while avoiding excessive distortion in highly deformed regions.

4.2 Structural Solver Settings

The structural solver employs a dynamic, non-linear approach to accurately capture the time-dependent deformation of the structure under aerodynamic loading. The structural response is governed by the implicit Newmark scheme, a widely used time integration method known for its balance between stability and accuracy in transient dynamic problems. The implicit nature of the scheme ensures numerical stability, even for large deformations.
A residual-based convergence criterion is used, ensuring high accuracy with strict tolerances of 10^{-6} for both displacement and residual norms. This guarantees that each time step achieves a physically meaningful solution before proceeding to the next iteration.
To handle the large, sparse matrices arising from finite element discretization, the solver employs a sparse QR linear solver, which is well-suited for structural mechanics simulations. This choice of solver enhances computational efficiency while maintaining robustness in solving complex mechanical systems.
The model constraints enforce full displacement fixation at the bottom boundary, replicating the physical clamping of the beam. This ensures that structural deformations only occur in the free regions of the beam, allowing an accurate assessment of its bending and oscillatory responses.
To evaluate structural response under aerodynamic loads, selected nodal displacements are monitored at specific locations. Displacement and reaction forces at critical points are recorded to assess the dynamic behavior and validate coupling interactions.

4.3 Fluid-Structure Coupling

The coupling strategy follows a Gauss-Seidel Strong Coupling approach, which ensures direct and iterative communication between the fluid and structural solvers within each time step. This method enhances the stability and accuracy of the simulation, particularly for strongly coupled FSI problems where bidirectional interactions are significant.
The coupling operates over 12 iterations per time step, ensuring that the fluid and structural solutions are sufficiently synchronized before advancing to the next step. This iterative exchange refines the interaction forces and displacements, reducing residual errors in the coupled system.
For efficient data transfer, the nearest neighbor mapping method is employed to exchange information between the fluid and structural domains. This approach is computationally efficient and ensures that nodal information is mapped correctly between the two meshes. Load transfer from the fluid to the structure is handled with sign swapping to ensure consistency in force application direction. A transpose operation is applied in the mapping of forces from fluid to structure, ensuring that reaction forces are correctly projected onto the structural nodes.
To improve convergence and computational efficiency, a Minimum Variance Quadratic Newton (MVQN) acceleration technique is implemented. This method enhances the iterative process by refining the data

exchange, reducing the number of coupling iterations required for convergence.
The bidirectional information exchange ensures that fluid forces influence structural deformation, which in turn affects the fluid domain through mesh movement. Consistent force application is maintained through sign swapping and transposition techniques, ensuring that forces and displacements are physically meaningful across solvers. Iterative refinement of the coupling variables reduces residual errors and ensures an accurate interaction resolution.
This FSI model integrates a robust combination of fluid dynamics, structural mechanics, and strong coupling techniques to accurately simulate the interaction between a flexible beam and surrounding fluid flow. The ALE formulation with a monolithic Navier-Stokes solver ensures numerical stability, while the implicit Newmark method in the structural solver provides accurate deformation predictions. The Gauss-Seidel strong coupling strategy with MVQN acceleration enhances convergence and bidirectional data exchange. These combined methodologies create a highly reliable simulation framework for analyzing fluid-structure interactions in engineering applications.

4.4 Optimization

As discussed earlier the optimization strategy in this study relies on the DE algorithm to iteratively adjust the structural material property, Young's modulus, to minimize the error between the computed and measured data. DE is a population-based optimization method particularly well-suited for solving nonlinear and non-differentiable problems, such as those encountered in FSI simulations. It operates by maintaining a population of candidate solutions, each representing a possible configuration of Young's modulus values. These candidates are iteratively refined using mutation, crossover, and selection operations to explore and exploit the solution space effectively.
The optimization strategy illustrated in the Figure 6 integrates material property adjustments with FSI framework. The process begins with initializing parameters and entering an iterative loop to refine Young's modulus values. For each iteration, the material file is updated, and the FSI simulation is executed to compute the
displacement and velocity at the sensor locations. The simulation results are compared with reference data to evaluate the objective function, guiding the optimization.
The workflow continues until the convergence criterion is met, indicating that the error between computed and measured values has been minimized. The flowchart emphasizes key steps, including material updates, simulation execution, and error evaluation, ensuring systematic refinement of the structural model. The optimization process in this study involves two approaches for improving the Young's modulus within the domain:

4.4.1 Optimization using Two Displacement Sensors

In the first approach, the optimization focuses solely on displacement measurements from two sensors placed at specific locations on the structure as shown in 4. By minimizing the difference between computed and measured displacements, this method successfully achieved optimal Young's modulus values within the specified bounds of 2.0 MPa to 2.5 MPa. This approach leveraged the relatively stable and direct relationship between displacement and material properties. As a result, the optimization converged efficiently, producing consistent and accurate results. The computed displacements closely matched the reference values, showcasing the robustness of the DE algorithm when applied to a simpler error function shown in equation 7 defined exclusively by displacement data.

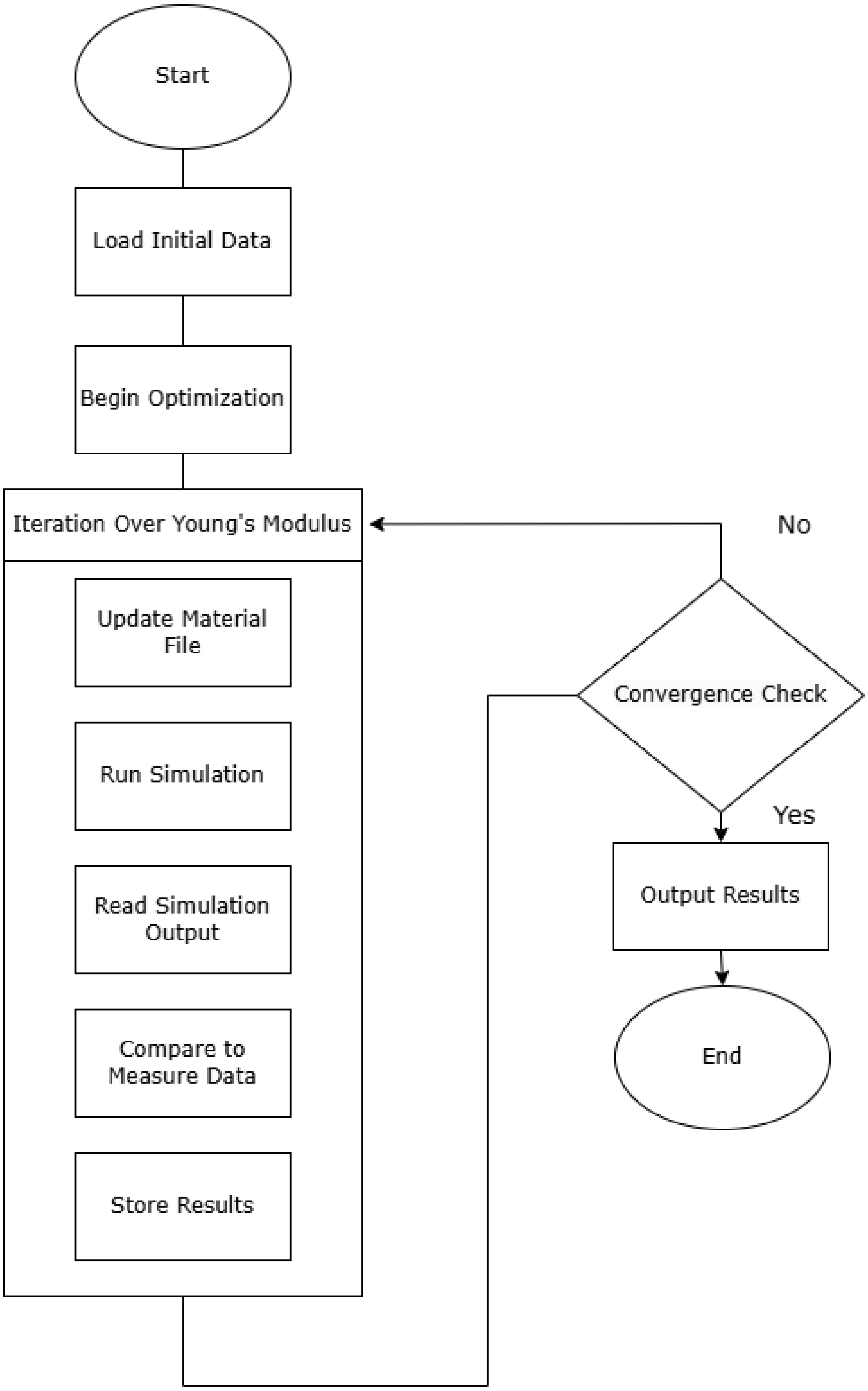

Figure 6: Optimization Strategy

4.4.2 Optimization using Two Displacement and Two Velocity Sensors

In the second approach, both displacement and velocity measurements were incorporated into the optimization process, aiming to provide a more comprehensive evaluation of the structural response. While this expanded error function added valuable information about the dynamic interaction between the structure and the fluid, it introduced challenges due to the unscaled velocity data. The velocity contributions, being significantly larger in magnitude compared to displacements, dominated the error function and skewed the optimization process. Consequently, the optimizer struggled to identify optimal Young's modulus values within the given bounds, often converging to suboptimal solutions. This issue highlights the critical importance of scaling the input data to ensure that all components of the error function contribute equitably. Without proper scaling, the optimizer prioritizes the velocity terms disproportionately, undermining its ability to balance the overall response.

5 Results

This section discusses the benchmarking, comparison of results and optimization process results.

5.1 Benchmarking & testing

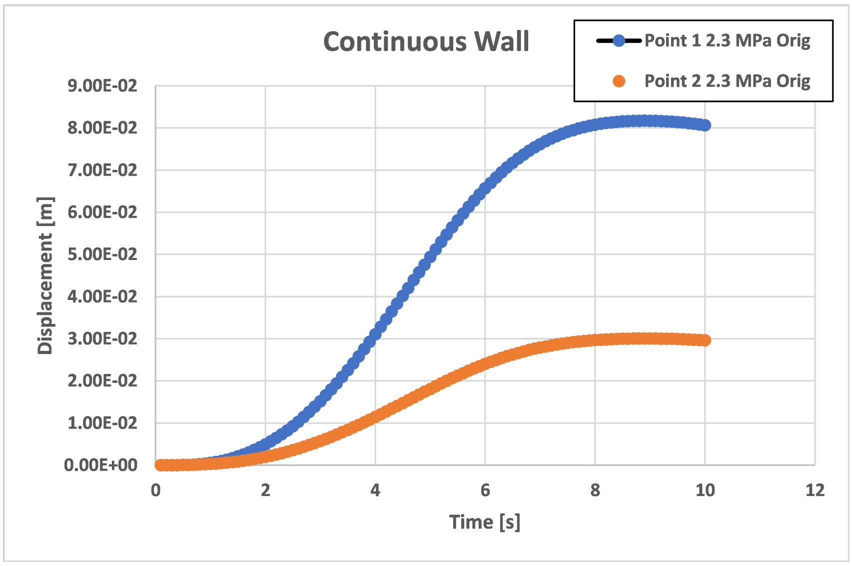

Figure 7: Original Wall without the Split

Figure 7 presents the displacement vs. time response for a continuous wall without a slit. This serves as the baseline case, providing insight into the natural deformation behavior of an unmodified structure under loading conditions. The displacement profile follows the expected trend, exhibiting smooth deformation characteristics.

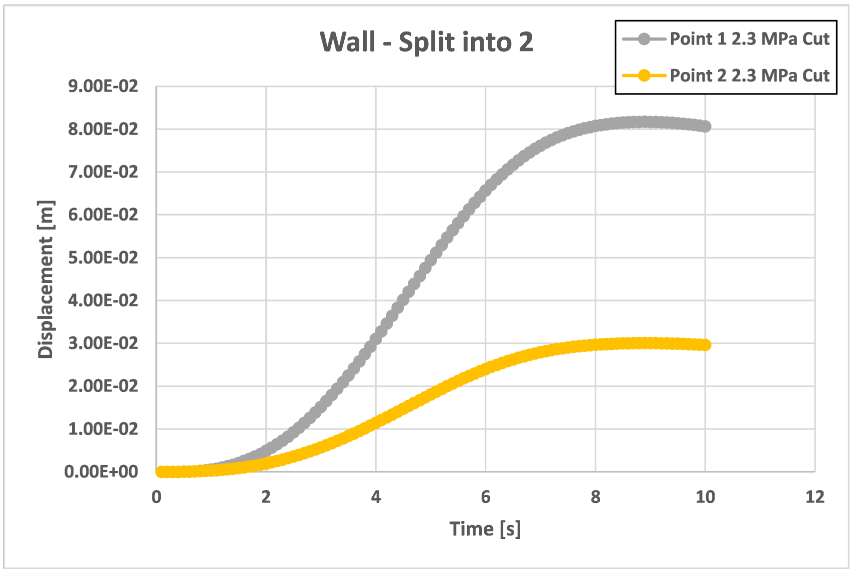

Figure 8: Wall Split into 2 halves

Figure 8 illustrates the displacement vs. time behavior for the same continuous wall after introducing a split. The presence of the slit alters the structural response, as discontinuities in the material can affect the propagation of displacement waves. This figure helps assess how the modification influences overall wall deformation compared to the unaltered case.

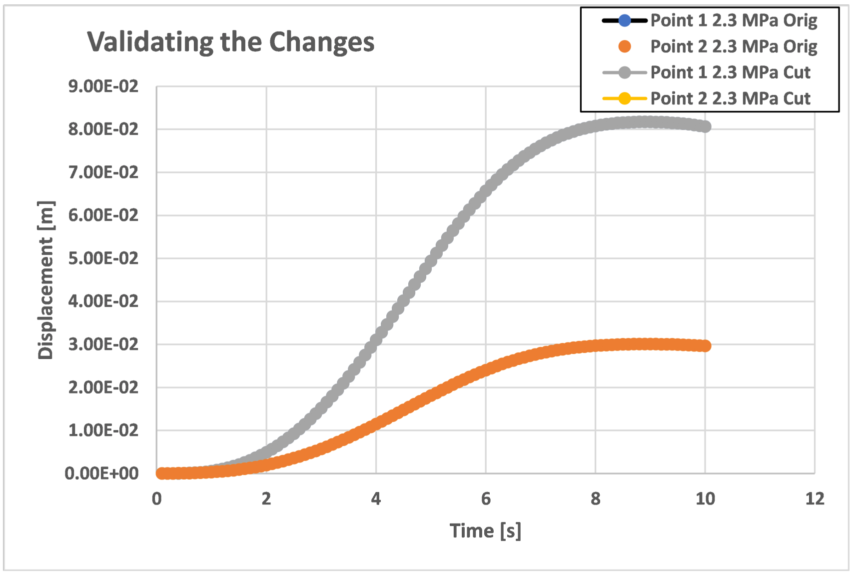

Figure 9: Comparing the Continuous and Split Wall

Figure 9 provides a direct comparison between the continuous and split-wall cases by plotting their displacement responses together. The near-perfect overlap of the displacement curves confirms that our mesh manipulation process was implemented correctly. This validates that the numerical simulation remains physically accurate and that the imposed split does not introduce unintended computational artifacts.

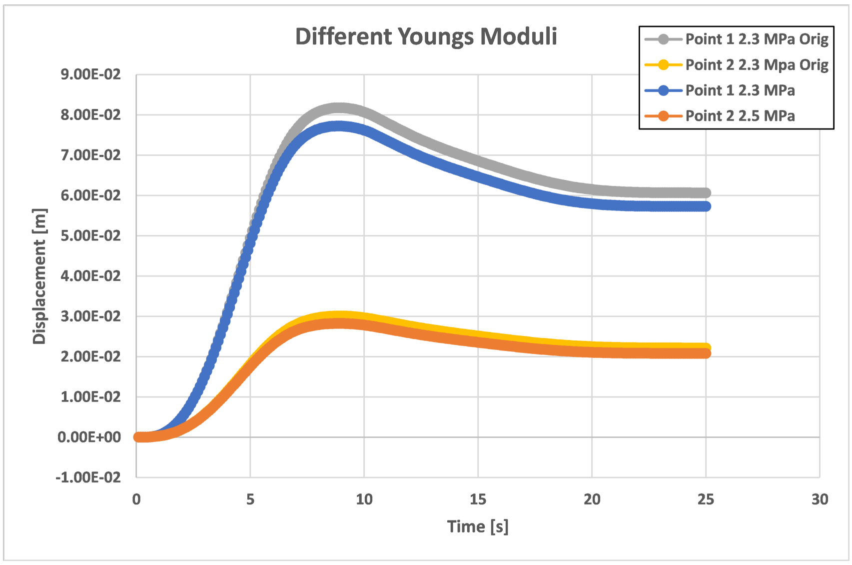

Figure 10: Bottom is Stiffer relative to the Top

Figures 10 and 11 further explore the effect of material stiffness variation within the split wall. In Figure 10, the bottom portion of the wall is assigned a higher Young's modulus than the top, making it significantly stiffer. This stiffness gradient influences the displacement distribution, restricting deformation in the lower section while allowing greater movement in the upper region.

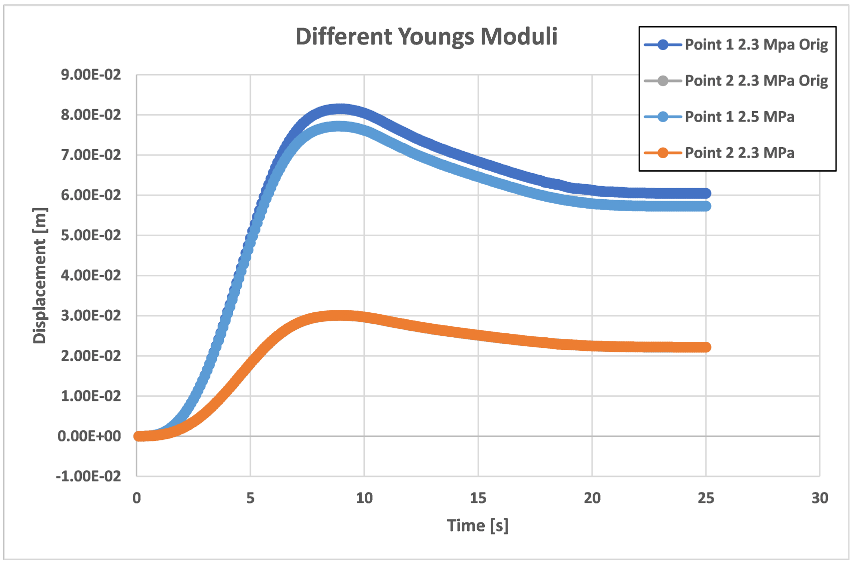

Figure 11: Top is Stiffer relative to the Bottom

Conversely, Figure 11 demonstrates the scenario where the top portion of the split wall is stiffer than the bottom. This reversal in material properties leads to a different deformation pattern, where the top section resists movement more effectively due to its increased stiffness, resulting in less displacement. On the other hand, the bottom section retains the original Young's modulus, leading to deformation that closely matches the behavior seen in the baseline case. As a result, the data line for the bottom section overlaps with the original displacement curve, while the top section shows reduced movement.

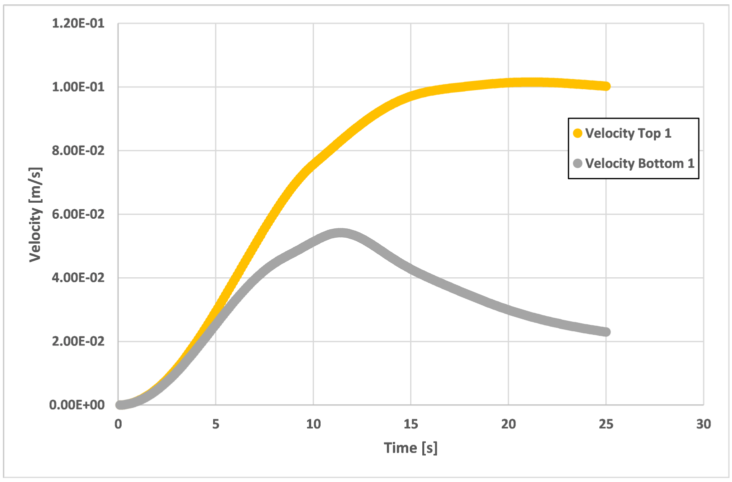

Figure 12: Velocity Measured by the Sensors

Figure 5 illustrates the placement of velocity sensors within the domain. One sensor is positioned in the region where vortices are initially generated, enabling the analysis of vortex shedding and localized flow disturbances. The other sensor is placed in the free stream at the top, providing a reference measurement of undisturbed flow conditions.

The ramp-up observed in Figure 12 is due to the cosine function commonly used in CFD simulations to smoothly establish the outlet pressure. This approach ensures numerical stability and prevents sudden pressure discontinuities in the computational domain.

The following two figures illustrate the flow simulation at different stages: one during the ramp-up phase Figure (13) and the other after the system has settled Figure 14. These visualizations help explain the variations in velocity measured by Sensor 2. As the beam deflects, a low-velocity channel forms at the top, causing fluctuations in the velocity recorded by Sensor 2 (highlighted in pink triangle). This explains why the measured velocity initially increases and then drops.

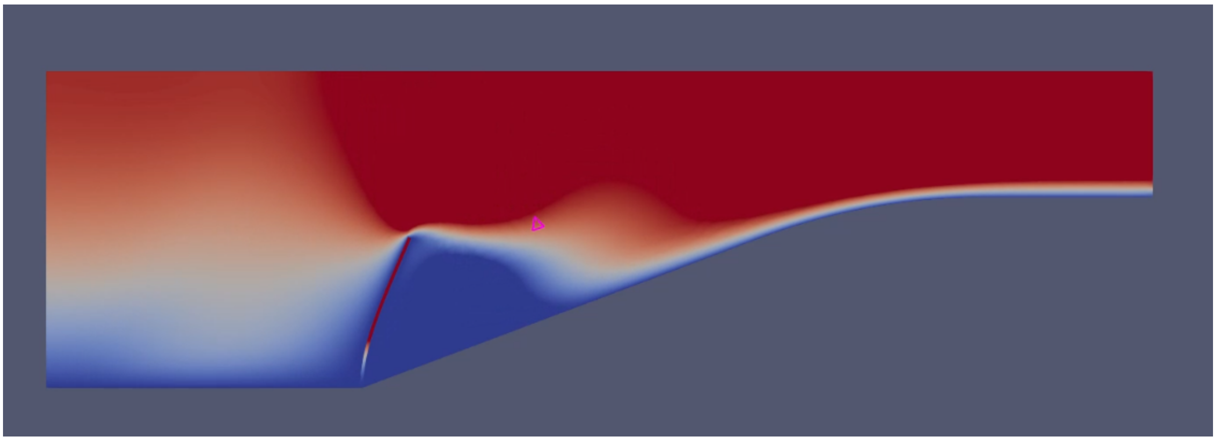

Figure 13: Velocity - During the Ramp-up

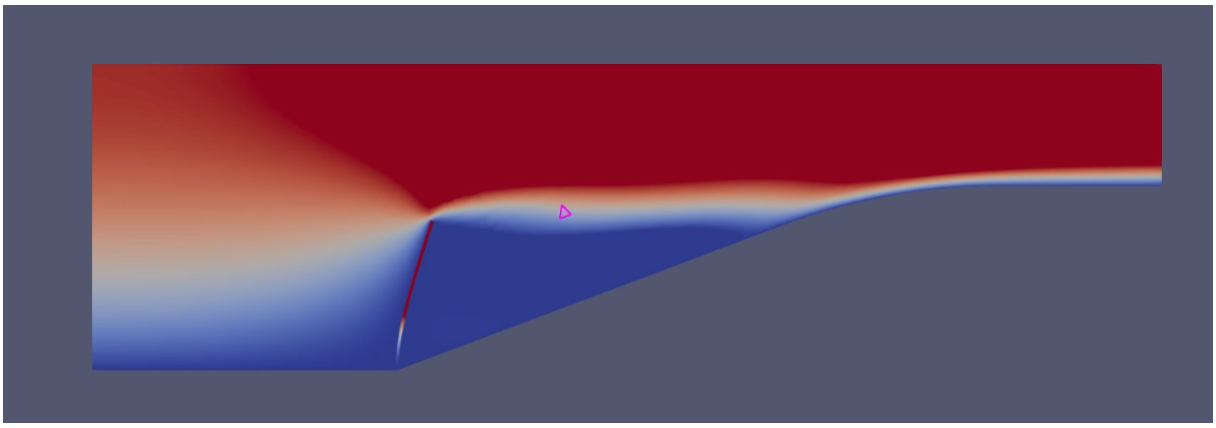

Figure 14: Velocity - After settling

In contrast, the velocity recorded by the other sensor remains constant, as it is located in the free stream, unaffected by localized disturbances near the beam.
These velocity measurements align with the flow visualizations, validating both the sensor placement and the accuracy of the recorded data. The consistency between the simulation results and measured values confirms the effectiveness of the velocity sensors in capturing the expected flow behavior.

5.2 Optimization Results

The optimization results shown in 15 for the four sensors show a rapid reduction in the objective function J from 5×10^{-3} to 2.5×10^{-3} within the first five iterations, followed by stabilization. This indicates that the DE algorithm effectively minimized the error between computed and measured data, achieving convergence efficiently. The remaining residual error suggests challenges such as unscaled velocity contributions and constrained Young's modulus bounds, which may limit further refinement. Despite this, the results highlight the robustness of the optimization framework, demonstrating its effectiveness for system identification in FSI simulations.

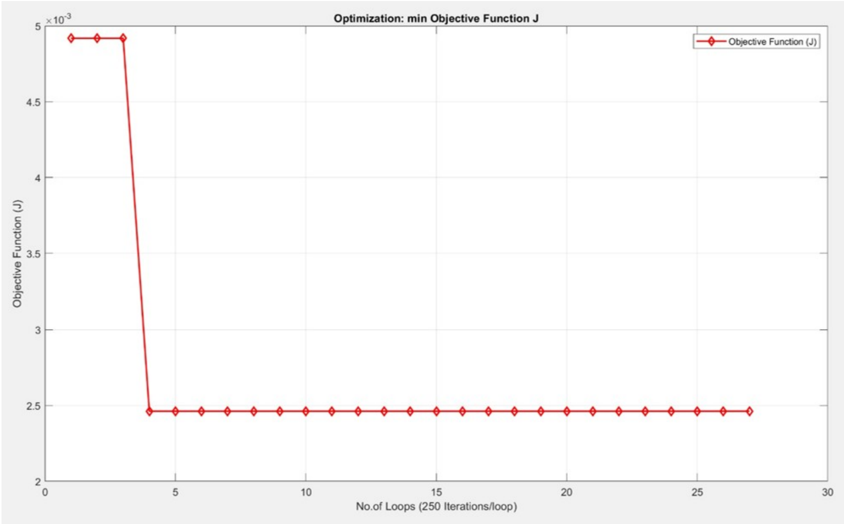

Figure 15: Optimization of function J

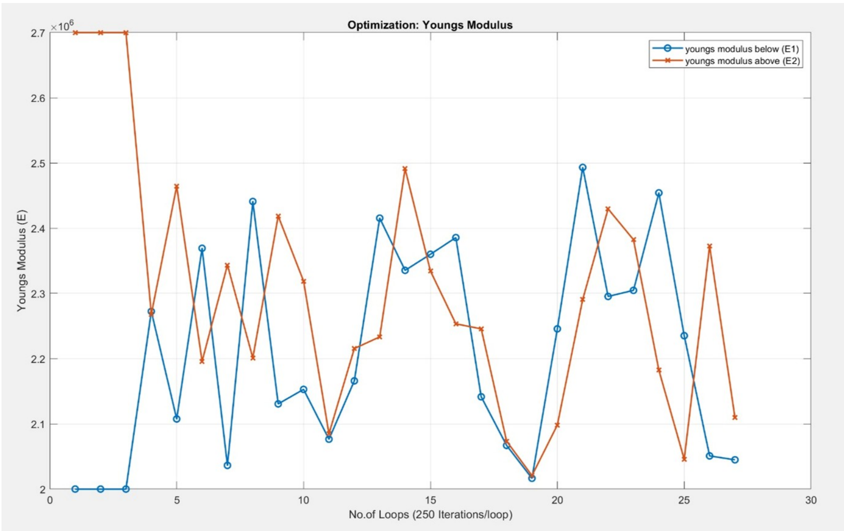

Figure 16: Optimization of Young's Modulus

The optimization process aimed to converge $E1$ and $E2$ toward the expected 2.3 MPa, but persistent oscillations indicate challenges in achieving stable refinement shown in Figure 16. Initial iterations show large fluctuations as the DE algorithm explores the solution space. Although the values approach 2.3 MPa in later iterations, unscaled velocity contributions in the error function probably dominate, disrupting precise convergence. Furthermore, the imposed limits may have restricted the search space of the algorithm. Despite these limitations, the framework effectively maintains realistic material properties and refines them iteratively.

5.3 Comparison of coupling strategies

The behavior of different coupling strategies—Gauss-Seidel strong, Gauss-Seidel weak, Jacobi weak, and the benchmark simulation shown in Figure 17 offers critical insights into their suitability for strongly coupled FSI problems. Gauss-Seidel methods rely on sequential updates, where the solution of one domain (e.g., fluid) is immediately used to inform the solution of the other domain (e.g., structure) within the same iteration. This iterative dependency inherently allows strong coupling to synchronize fluid and structural

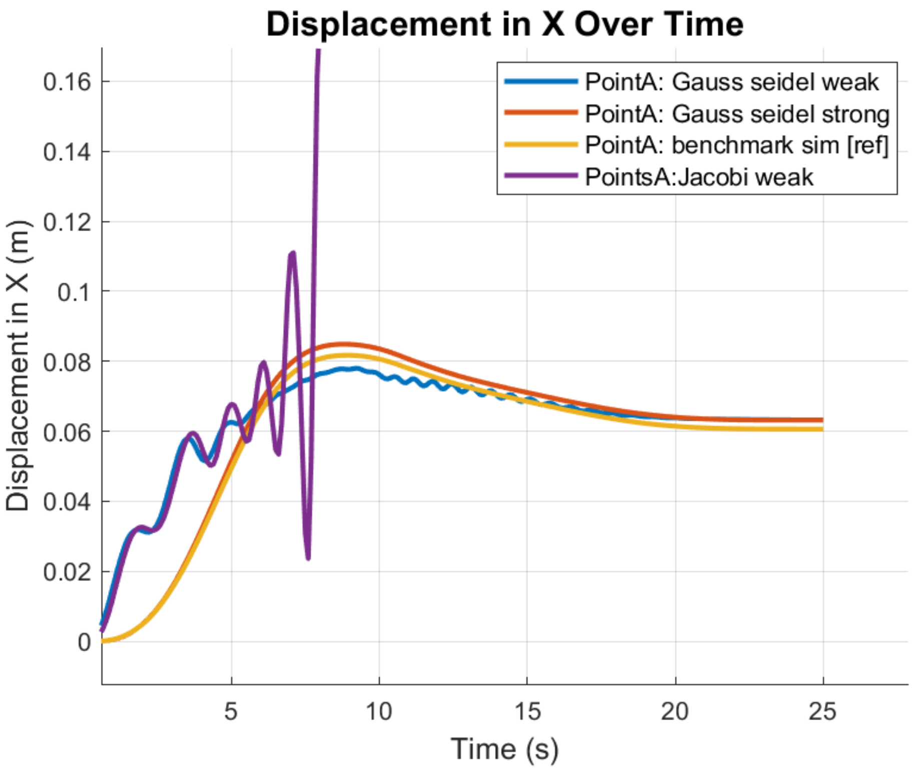

Figure 17: Coupling Strategy

responses effectively, as seen in the Gauss-Seidel strong strategy. Here, displacement results closely match the benchmark, with minimal deviation and faster convergence due to the efficient exchange of data between domains. On the other hand, Gauss-Seidel weak coupling, while stable and computationally cheaper, lacks the iterative dependency required for capturing highly nonlinear interactions in strong coupling scenarios. As a result, its displacement behavior converges more slowly but remains consistent with the benchmark over time.

Conversely, Jacobi methods rely on parallel updates where the solutions for both domains are computed independently within the same iteration, only exchanging data at the end. This independence makes the method inherently less stable in strongly coupled problems, as seen in the Jacobi weak strategy, where pronounced oscillations and divergence occur. The failure to synchronize fluid and structural dynamics effectively underpins this behavior, as forces and displacements exchanged between the domains may not align in direction or magnitude, leading to spurious results. This is particularly evident in highly interactive systems, where the inability of Jacobi weak to resolve bidirectional feedback exacerbates instabilities, resulting in large spikes and oscillations in the displacement graph. In contrast, Gauss-Seidel strong coupling iteratively resolves these interactions, ensuring consistent updates and convergence. Under the hood, this reflects how Gauss-Seidel's sequential dependency aligns the interaction forces and displacements across the coupled domains, while Jacobi's independence disrupts this synchronization, making it unsuitable for strongly coupled problems.

6 Conclusion and Future Work

This study presents a robust and adaptive framework for FSI simulations coupled with optimization strategies, demonstrating its potential for addressing complex structural optimization challenges. Through the development and analysis of an automated beam segmentation and optimization process, the study successfully highlighted the interplay between material property distribution and structural performance within a dynamic fluid environment. The methodology's ability to handle both equal and unequal segmentation with optimal Young's modulus values underscores its versatility and reliability in generating targeted design modifications. By leveraging optimization techniques, the framework minimized beam deflection efficiently while preserving the integrity of the structural simulation, providing a seamless balance between accuracy and computational feasibility.

A detailed analysis of coupling strategies revealed the significance of choosing appropriate solvers in FSI problems. The Gauss-Seidel strong coupling method emerged as the most reliable approach for ensuring accurate synchronization between fluid and structural domains, closely replicating benchmark simulations with faster convergence. In contrast, weak coupling methods demonstrated slower convergence, with the Jacobi weak strategy exhibiting significant oscillations due to its parallel update mechanism, which is ill-suited for strongly coupled systems. These findings emphasize the importance of solver selection in capturing bidirectional interactions and achieving stable, accurate solutions. Additionally, the implementation of a Gaussian weak coupling strategy during optimization enabled computational efficiency, although it introduced minor fluctuations in the error function due to reduced synchronization between domains. Nevertheless, the tool chain maintained robustness, enabling effective material property optimization within acceptable computational limits.

Future work will focus on expanding the framework's scalability and automation. Incorporating automated input handling for beam segmentation ratios and Young's modulus distributions will further streamline the optimization process. Enhancing the error function approximation, particularly for systems with high sensitivity to fluid-structure interactions, will improve solution stability and accuracy. Moreover, extending the methodology to handle three-dimensional structures and complex geometries will unlock new possibilities for applications in aerospace, biomechanics, and other engineering fields. Integrating adaptive meshing techniques and advanced preconditioning strategies will address computational bottlenecks, ensuring the framework's applicability to larger and more intricate simulations. Ultimately, the insights gained from this study lay the groundwork for advancing the capabilities of FSI simulations and their integration into real-world structural optimization problems.

This study concludes with a reaffirmation of the importance of solver accuracy, optimization adaptability, and computational efficiency in achieving reliable solutions for strongly coupled systems. The developed framework serves as a stepping stone toward more sophisticated methodologies, fostering further innovation in the field of FSI modeling and optimization.

References

[1] Kratos Multiphysics Developers.
CoSimulationApplication Overview - Kratos Multiphysics.
Accessed: 2025-01-31.
2025.
URL: `https://tinyurl.com/CoSimulationApplication`.